3.2 Understanding and Interpreting Confidence Intervals

- We estimate parameters by using "confidence intervals" that are derived from the sampling distributions.
- This section focuses on
  1. Interval estimate
  2. Interval estimate based on margin of error
  3. Confidence interval
  4. 95% confidence interval using the SE
  5. Understanding confidence intervals
  6. Common misinterpretations

Statistical Inference

- A population is characterized by unknown numerical descriptive measures called **parameters**.
- Typical parameters of interest are
  1. a population mean μ
  2. a population proportion p
- Statistics makes inferences about these unknown parameters based on information contained in a sample.
- Two categories of statistical inference:
  1. **Estimating** an unknown parameter answers "*What is the true value of the parameter*?"
  2. **Testing** a hypothesis about an unknown parameter answers "*Is the true value of parameter equal to the specific value in the hypothesis*?"

Point Estimation

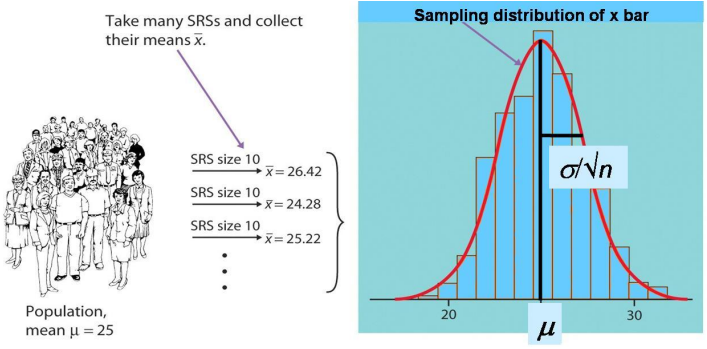
- A **point estimate** of an unknown parameter is a single value calculated from a sample data to estimate the parameter.
- A point estimate is called an **unbiased estimate** if its sampling distribution has a mean value equal to the true value of the parameter.
- Example:
  1. $\bar{x}$ is a point estimate of μ.
  2. $\bar{x}$ is an unbiased estimate of μ because the mean value of the sampling distribution of $\bar{x}$ is equal to the population mean. Recall $\mu_{\bar{x}} = \mu$.
  3. Therefore, we expect $\bar{x}$ to be "close" to μ.
  4. However, an observed $\bar{x}$ itself is a single value so it is almost certainly not correct for estimating μ.
  5. The closeness is measured in **standard error**, or standard deviation, of the $\bar{x}$-distribution. Recall $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

Sampling Distribution | Review

- Suppose all possible samples of a fixed sample size n are taken from a population and a sample statistic of interest, such as $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, is calculated from each sample.
- The **sampling distribution** of the statistic is the distribution of all possible values of the sample statistic.
- Statistical inference is based on the sampling distribution.

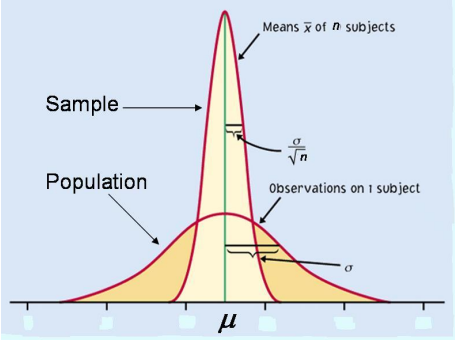
Sampling Distribution of $\bar{x}$ | Sketch

- Suppose all possible samples of a fixed sample size n are taken from a population with mean μ and variance σ^2 and $\bar{X}$ is calculated from each sample.
  1. Some $\bar{x}$'s will be greater than the population mean μ and some $\bar{x}$'s will be smaller, making up the sampling distribution.
  2. These $\bar{x}$'s have a normal distribution centered at μ with standard deviation $\frac{\sigma}{\sqrt{n}}$.



Sampling distribution of $\bar{x}$ | Illustration

- Recall the $\bar{x}$-distribution is narrower than the population distribution, by a factor of $\sqrt{n}$.
- Therefore, the values of $\bar{x}$ calculated from different samples are relatively close to the population mean μ.



If the population is $N(\mu, \sigma)$, then the $\bar{x}$-distribution is $N(\mu, \frac{\sigma}{\sqrt{n}})$.

Sampling Distribution of $\bar{x}$ | Theory

- Suppose $x_1, \ldots, x_n$ is a random sample of size n from a population with mean μ and standard deviation σ.
- The sampling distribution of the sample mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ has
  1. a normal distribution if n is sufficiently large
  2. theoretical mean $\mu_{\bar{x}}$ equal to the population mean μ: $\boxed{\mu_{\bar{x}} = \mu}$
  3. theoretical standard deviation $\sigma_{\bar{x}}$ equal to the population standard deviation σ divided by $\sqrt{n}$: $\boxed{\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}}$
- Notes
  1. In practice, a sample of size n is taken only once, and only one value of the sample statistic is calculated.
  2. A sampling distribution is therefore not an empirically observed distribution. Rather, it is a theoretical concept. In practice, we do not build it.

Sampling Distribution of $\bar{x}$ | Simulation

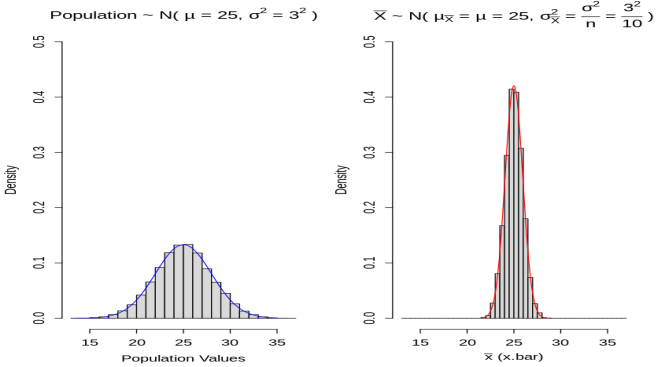
- R scripts to generate a sampling distribution of $\bar{x}$:

```r
mu = 25;  sigma = 3                      # population mean and standard deviation
n = 10;                                  # sample size
n.rpt = 5000                             # number of repetitions

x.bar = numeric( length=n.rpt )          # an empty vector for n.rpt sample means
set.seed( 12345 )                        # a seed value for random number generation
                                         # to reproduce the same result
for ( k in 1:n.rpt ) {                   # repeat the following procedures n.rpt times
  x.smp = rnorm( n, mean=mu, sd=sigma ) # generate a sample from N(mu,sigma^2)
  x.bar[k] = mean( x.smp )               # compute & keep the sample mean for each sample
}
```

Population Distribution vs. Sampling Distribution of $\bar{x}$

- Population distribution (left) vs. Simulated sampling distribution of $\bar{x}$ (right)



- We use this $\bar{x}$-distribution to perform statistical inference on the population mean μ.

Example 3.1 | Sampling distribution of the sample mean

- The weight in grams of single brown eggs is normally distributed with mean $\mu = 65$ and standard deviation $\sigma = 5$. Think of a carton of 12 eggs as a random sample.

  1. What are the mean and standard deviation, or standard error (SE), of the sampling distribution of the sample mean $\bar{x}$?

     Answer. The mean of the $\bar{x}$-distribution is

     $$\mu_{\bar{x}} = \mu = 65.$$

     The standard deviation, or standard error (SE), of the $\bar{x}$-distribution is

     $$\text{SE} = \frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{12}} = 1.44.$$

  2. Find the middle 95% of the sample mean distribution.

     Answer. Roughly ± 2 standard deviations from the mean $\mu = 65$. That is,

     $$\mu_{\bar{x}} \pm (2)(\text{SE}) = 65 \pm (2)(1.44) = 65 \pm 2.88 = (62.12, 67.88)$$

Sampling Distribution of $\hat{p}$ | Theory

- Suppose a random sample of size n is selected from a population with proportion p.
- The sampling distribution of the sample proportion $\hat{p} = \dfrac{\text{number in category}}{\text{total number}}$ has
  1. a normal distribution if n is sufficiently large
  2. theoretical mean $\mu_{\hat{p}}$ equal to the population proportion p: $\boxed{\mu_{\hat{p}} = p}$
  3. theoretical standard deviation $\sigma_{\hat{p}}$ equal to $\sqrt{\frac{p(1-p)}{n}}$: $\boxed{\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}}$

Example 3.2 | Sampling distribution of the sample proportion

- A survey of 500 adults finds that 60% support a proposed health policy. Suppose the true population proportion of adults who support the policy is $p = 0.58$.

1. What are the mean and standard deviation, or standard error (SE), of the sampling distribution of the sample proportion $\hat{p}$?

   Answer. The mean of the $\hat{p}$-distribution is

   $$\mu_{\hat{p}} = p = 0.58.$$

   The standard deviation, or standard error (SE), of the $\hat{p}$-distribution is

   $$\text{SE} = \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.58(1-0.58)}{500}} = \sqrt{0.000487} = 0.022.$$

2. Find the middle 95% of the sample proportion distribution.

   Answer. We find

   $$\mu_{\hat{p}} \pm (2)(\text{SE}) = 0.58 \pm (2)(0.022) = 0.58 \pm 0.044 = (0.536, 0.624)$$

Interval Estimation

- Consider an interval for an unknown parameter:

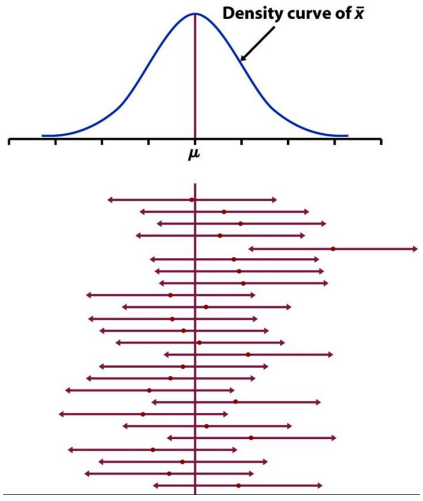
$$\text{point estimate} \pm \text{margin of error}$$

- This interval will contain the true value of the parameter a certain proportion of the time. This proportion, called the **confidence level**, represents how confident we are that the interval will capture the true value of parameter.
- Notes: A 95% confidence interval has a confidence level of 0.95.
  1. We are 95% confident that the method we use to construct this interval will capture the true value of parameter.
  2. The other interpretation is that 95% of all possible intervals we could construct in repeated sampling will contain the true value of parameter. The other 5% will not.

Statistical Confidence

- **95% of all sample means will be within roughly 2 standard deviations ($2 \times \frac{\sigma}{\sqrt{n}}$) of the population mean μ.**
- Equivalently, because distances are symmetrical, **the population mean μ must be within roughly 2 standard deviations from the sample mean, in 95% of all samples**.

(Red dot: the mean value of an individual sample)



Confidence Interval for Population Mean

- The **confidence interval** is a range of values with an associated probability or **confidence level** C. The probability quantifies the chance that the interval contains the true population parameter.
- A 95% confidence interval for the population mean μ is

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

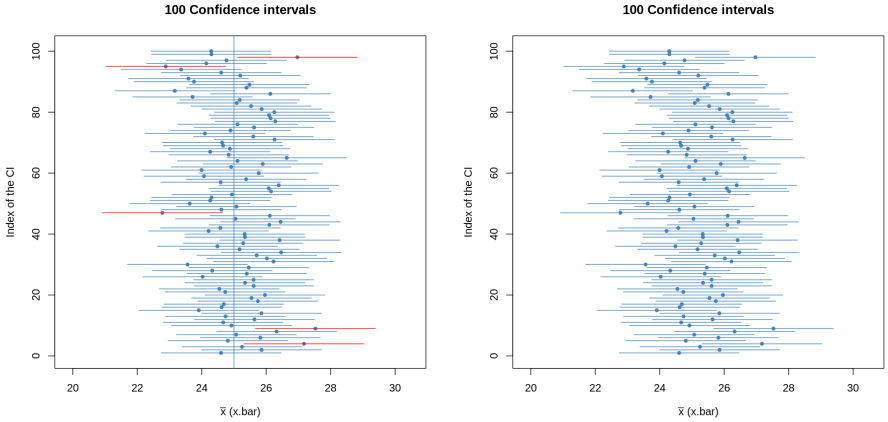
1. Why? Because $\bar{X} \sim \text{N}(\mu_{\bar{X}} = \mu, \sigma^2_{\bar{X}} = \frac{\sigma^2}{n})$, we have

$$\boxed{P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = 0.95} \Rightarrow \boxed{P\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95}$$

2. We assumed σ is known.
3. Otherwise, we will replace σ with the sample standard deviation s and 1.96 with an equivalent quantile from a t-distribution with $n - 1$ degrees of freedom (if n is small or moderate).

Confidence Interval for Population Mean

- Confidence intervals: $\mu = 25$ is known (left) vs. μ is unknown (right)



1. If μ is known (left), 95% of these CIs contain μ, and in addition, we know CIs that contain μ.
2. In practice, μ is unknown (right). We cannot tell whether a particular CI contains unknown μ or not. However, the percentage of CIs that contain unknown μ remains the same: 95%.

Popular Confidence Intervals for Population Mean

- Three most popular confidence intervals for μ are
  1. 90% confidence interval: $\bar{x} \pm (1.645)(\text{SE})$ or $\bar{x} \pm 1.645\frac{\sigma}{\sqrt{n}}$
  2. 95% confidence interval: $\bar{x} \pm (1.96)(\text{SE})$ or $\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$
  3. 99% confidence interval: $\bar{x} \pm (2.576)(\text{SE})$ or $\bar{x} \pm 2.576\frac{\sigma}{\sqrt{n}}$
- The condition that the population standard deviation σ is known is not realistic. We will discuss in Chapter 6 about the confidence interval estimation method for μ when σ is unknown.

Interpreting a Confidence Interval

- 95% of all samples yield intervals that contain the true parameter, so we say we are "**95% sure**" or "**95% confident**" that one interval contains the true parameter.
- For example, "We are **95% confident** that the **true proportion** of all Americans that considered the economy a 'top priority' in January 2012 is **between 0.84 and 0.88**."

Common Misinterpretations of CIs

- Some common misinterpretations of CIs
  1. "A 95% confidence interval contains 95% of the data in the population."
     Why?
  2. "I am 95% sure that the mean of a sample will fall within a 95% confidence interval for the mean."
     Why?
  3. "The probability that the population parameter is in this particular 95% confidence interval is 0.95."
     Why?

Implications for Practice

- We do not need to take a lot of random samples to "rebuild" the sampling distribution and find the true population mean μ, or the true population proportion p, at its center.
- All we need is one random sample of size n, and we rely on the properties of the sampling distribution to infer the true population mean μ, or the true population proportion p.

Example 3.3 | Confidence Interval for Population Mean

- A consumer buys a carton of 12 white eggs. These 12 white eggs weigh 770 g. Assume the standard deviation of white egg weights is 5 g. What can you infer about the mean weight of the white egg population?

  Answer. The average egg weight from this sample is $\bar{x} = \frac{770}{12} = 64.2$. We are 95% confident that the population mean weight μ is roughly within 2 standard deviations of $\bar{x}$. Specifically, the 95% CI for μ is

$$\bar{x} \pm (2)(\text{SE}) = \bar{x} \pm 2\frac{\sigma}{\sqrt{n}} = 64.2 \pm 2\frac{5}{\sqrt{12}} = 64.2 \pm (2)(1.44) = 64.2 \pm 2.88 = (61.32, 67.08)$$

Example 3.4 | Confidence Interval for Population Mean

- A study wants to determine the average number of hours, μ, of day care used in the last month. A random sample of $n = 36$ families produces $\bar{x} = 138$ hours. Assume the standard deviation for day care in hours is 18.
  1. Find a 95% confidence interval for μ.

     Answer. With $\sigma = 18$, we obtain a 95% confidence interval for μ by

     $$\bar{x} \pm (1.96)(\text{SE}) = \bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}} = 138 \pm 1.96\frac{18}{\sqrt{36}} = 138 \pm 5.88 = (132.12, 143.88).$$

     Therefore, we are 95% confident that the mean number of hours of day care used per month is between 132.12 and 143.88.
  2. Find a 99% confidence interval for μ.

     Answer. A 99% confidence interval for μ is

     $$\bar{x} \pm (2.576)(\text{SE}) = \bar{x} \pm 2.576\frac{\sigma}{\sqrt{n}} = 138 \pm 2.576\frac{18}{\sqrt{36}} = 138 \pm 7.728 = (130.272, 145.728)$$

     Therefore, we are 99% confident that the families spent on average 130.27 to 145.73 hours for day care in the last month.

Exercise 3.1

- A 95% confidence interval for the mean μ of a population is computed from a random sample and found to be 9 ± 3. We may conclude
  1. there is a 95% probability that μ is between 6 and 12.
  2. there is a 95% probability that the true mean is 9, and there is a 95% chance that the true margin of error is 3.
  3. if we took many, many additional random samples and from each computed a 95% confidence interval for μ, approximately 95% of these intervals would contain μ.
  4. all of the above.
- Suppose the population of the scores of all high school seniors who took the SAT math test this year follows a normal distribution with mean μ and standard deviation $\sigma = 100$. A report that says, "On the basis of a random sample of 100 high school seniors who took the SAT math test this year, a confidence interval for μ is 512 ± 25.76." The confidence level is
  1. 90%
  2. 95%
  3. 99%
  4. more than 99.9%

Exercise 3.2

- Suppose the weights of boxes of cereal are normally distributed with a standard deviation of 0.29 ounces. A random sample of 25 boxes had a mean weight of 9.82 ounces.
  1. Compute a 90% confidence interval for the mean weight of the cereal boxes.

  2. Compute a 95% confidence interval for the mean weight of the cereal boxes.

Exercise 3.3

- A survey of 1,502 Americans in January 2012 found that 86% consider the economy a “top priority” for the president and congress. The standard error (SE) for this statistic is 0.01. What is the 95% confidence interval for the true proportion of all Americans that considered the economy a “top priority” at that time?